Predicting Growth and Remodeling of Cardiovascular Tissue

change in mass

change in microstructure

Dr.-Ing. Martin R. Pfaller

Assistant Professor

Department of Biomedical Engineering, Yale University

`biomechanics.yale.edu`

With much inspiration from Drs. Amadeus Gebauer, Jonas Eichinger, and Jay Humphrey

Student Version

Yale

The Origin of Mechanobiology

Davis' law, **Henry Gassett Davis** (1807–1896, USA), 1867:

"Ligaments, or any soft tissue, when put under even a moderate degree of tension, if that tension is unremitting, will elongate by the addition of new material."

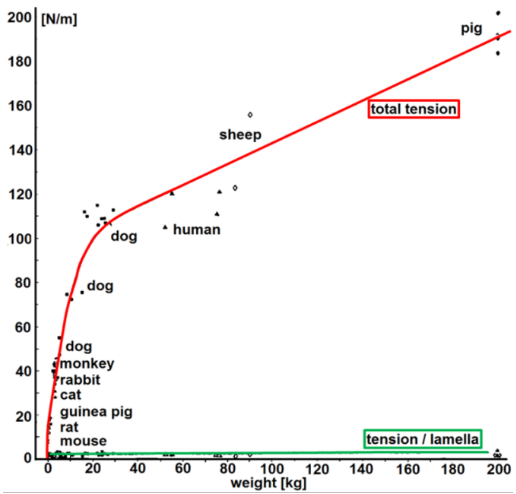
Wolff's law, **Julius Wolff** (1836–1902, Germany), 1892 (*Das Gesetz der Transformation der Knochen*, the law of bone transformation):

"im Gefolge primärer Abänderungen der Form und Inanspruchnahme oder auch bloß der Inanspruchnahme der Knochen [sich] bestimmte, nach mathematischen Regeln eintretende Umwandlungen der inneren Architektur und ebenso bestimmte, denselben mathematischen Regeln folgende sekundäre Umwandlungen der äußeren Form der betreffenden Knochen vollziehen."

- bone mass proportional to load (higher / lower load $\Rightarrow$ growth / atrophy)
- first *mathematical* relation between load and growth & remodeling in biological tissue

A Brief Introduction to Mechanobiology of Blood Vessels

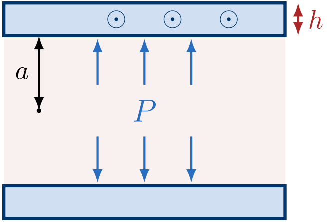
Vascular tissue has a preferred mechanical state



Tension T varies by a **factor of 26**

$$T = P\,a$$

T and σ_θ

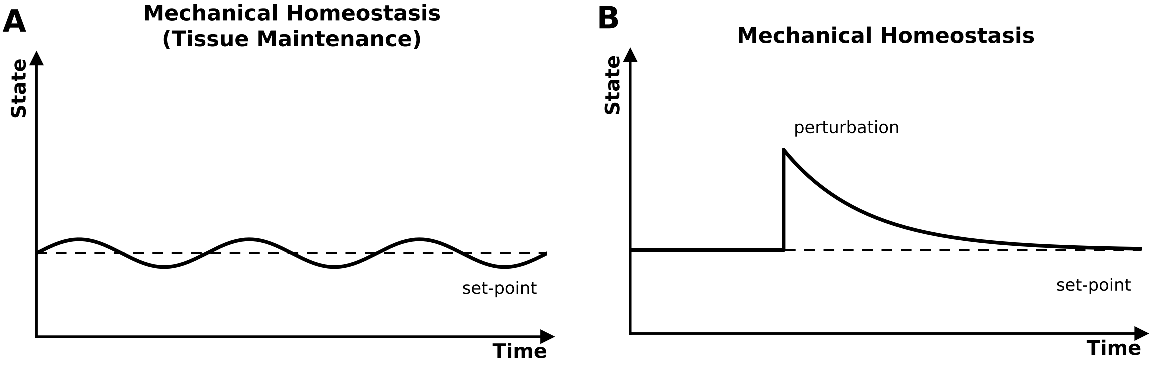


Tension / lamella varies by a **factor of 2.8**

$$\frac{T}{h} = \frac{P\,a}{h} = \sigma_\theta$$

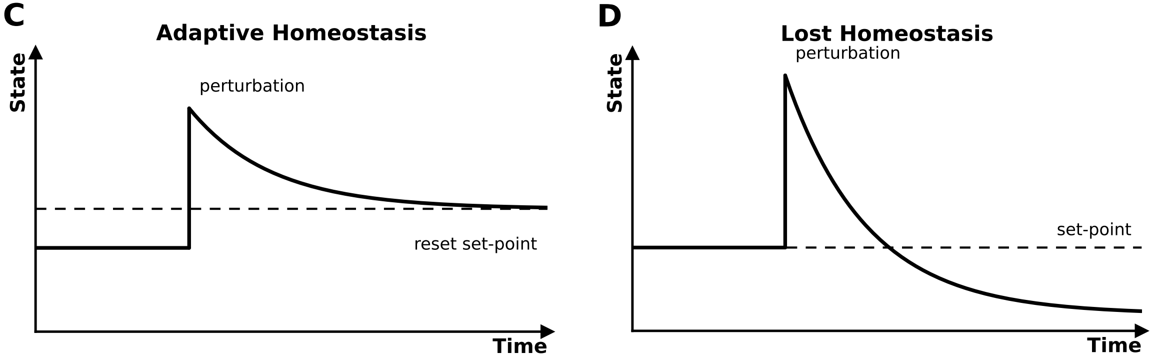
Wolinsky & Glagov, Circ Res (1967); Fig. 8 adapted by J. Eichinger

Mechanical Homeostasis



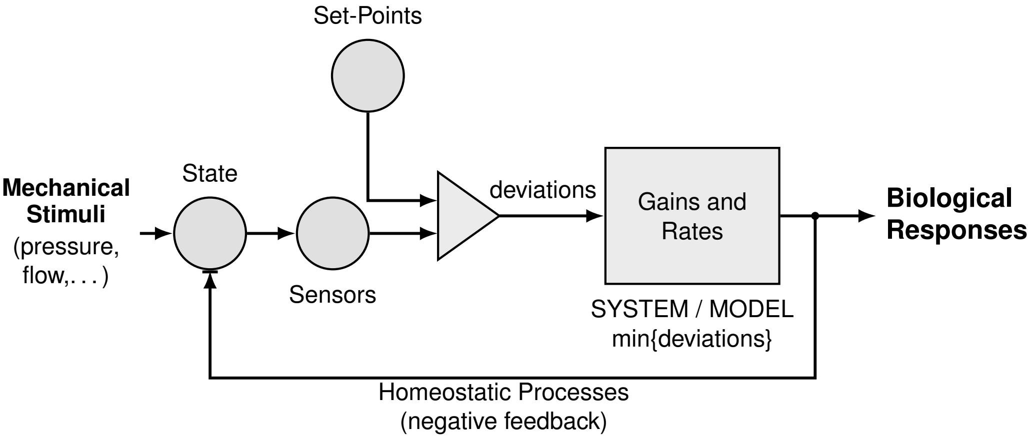
Humphrey & Schwartz, Annu Rev Biomed Eng (2021)

(Mal)adaptation



Humphrey & Schwartz, Annu Rev Biomed Eng (2021)

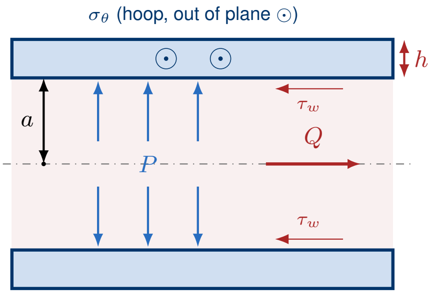
Mechanobiological Regulation



Humphrey & Schwartz, Annu Rev Biomed Eng (2021)

An Idealized Blood Vessel

A vessel is a **pressurised, perfused tube**. Two loads act on the wall: **blood pressure** P pushes it outward, and **blood flow** Q drags its inner surface. Cells sense both.



a inner radius, h wall thickness
P lumen pressure $\rightarrow$ **circumferential stress** $\sigma_\theta \sim 100\,$kPa
Q volumetric flow $\rightarrow$ **wall shear stress** $\tau_w \sim 1\,$Pa

Two Loads, Two Formulae, Two Adaptations

Pressure P sets the **hoop stress**; flow Q sets the **wall shear stress**. Each has a set-point; rearranging shows the geometry that restores it.

Pressure (Laplace)

$$\sigma_\theta = \frac{Pa}{h} \tag{1}$$

Flow (Poiseuille)

$$\tau_w = \frac{4\mu Q}{\pi\, a^3} \tag{2}$$

Laplace & Poiseuille; Humphrey, Cardiovascular Solid Mechanics (2002)

Your turn: Pressure adaptation

Question. Pressure **doubles**. How does the wall thickness change? $\sigma_\theta = \dfrac{Pa}{h}$

Hu, ..., Humphrey, Ann Biomed Eng (2006)

Your turn: Pressure adaptation

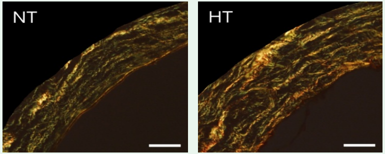
Question. Pressure **doubles**. How does the wall thickness change? $\sigma_\theta = \dfrac{Pa}{h}$

Answer: Pressure adaptation

$P'/h' = P/h$ with $P' = 2P$ gives

$$h' = \left(\frac{P'}{P}\right) h = 2\,h. \tag{3}$$

Wall thickens *in proportion* to pressure.



$P \uparrow$

Your turn: Flow adaptation

Question. Flow rises **8-fold**. How does the radius change? $\tau_w = \dfrac{4\mu Q}{\pi\, a^3}$

Your turn: Flow adaptation

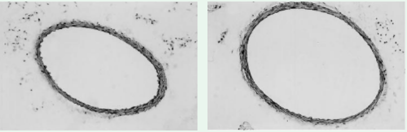
Question. Flow rises **8-fold**. How does the radius change? $\tau_w = \dfrac{4\mu Q}{\pi\, a^3}$

Answer: Flow adaptation

$(a'/a)^3 = Q'/Q = 8$, so

$$a' = \left(\frac{Q'}{Q}\right)^{1/3} a = 2\,a. \qquad (4)$$

Reading. Radius grows as the *cube root* of flow.



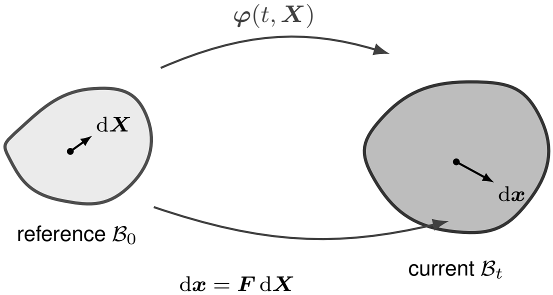
$Q \uparrow$

Finite-Strain Fundamentals

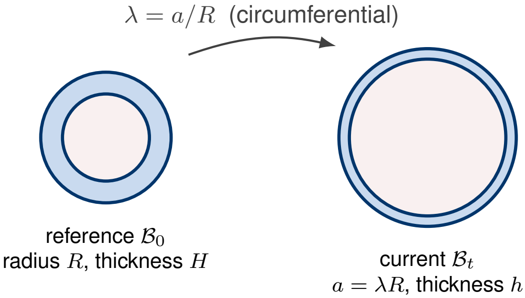
Reference and Current Configuration

A body is described by a **reference** $\mathcal{B}_0$ (points $\boldsymbol{X}$) and a **current** $\mathcal{B}_t$ (points $\boldsymbol{x}$) configuration, related by the motion $\boldsymbol{x} = \boldsymbol{\varphi}(t, \boldsymbol{X})$.

3D 3D geometric model



0D 0D reduced model



The Deformation Gradient

The **deformation gradient** maps reference line elements to current ones.

3D a 3×3 tensor

$$\mathrm{d}\boldsymbol{x} = \boldsymbol{F}\,\mathrm{d}\boldsymbol{X}, \qquad \boldsymbol{F} = \frac{\partial \boldsymbol{\varphi}}{\partial \boldsymbol{X}}, \qquad J = \det \boldsymbol{F} \tag{5}$$

J is the volume ratio, **incompressibility** means $J = 1$.

0D a scalar

$$\boldsymbol{F} \longrightarrow \lambda = \frac{\text{current length}}{\text{reference length}}, \qquad \lambda = 1 \text{ undeformed.}$$

Strain: A Rotation-Free Measure of Stretch

Removing rigid rotation gives the **right Cauchy–Green** tensor and the **Green–Lagrange strain**.

3D symmetric 3×3 tensors

$$\boldsymbol{C} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} = (\boldsymbol{R}\boldsymbol{U})^{\mathrm{T}}\boldsymbol{R}\boldsymbol{U} = \boldsymbol{U}^{\mathrm{T}}\boldsymbol{R}^{\mathrm{T}}\boldsymbol{R}^{\mathrm{T}}\boldsymbol{U} = \boldsymbol{U}^2, \qquad \boldsymbol{E} = \tfrac{1}{2}(\boldsymbol{C} - \boldsymbol{I}) \tag{6}$$

0D scalars

$$C = \lambda^2, \qquad E = \tfrac{1}{2}(\lambda^2 - 1).$$

Stress: "Force per Area"

Cauchy stress: *current* force per unit *current* area.

3D spatial symmetric 3×3 tensor (from 2nd Piola–Kirchhoff stress $\boldsymbol{S}$)

$$\boldsymbol{\sigma} = \tfrac{1}{J}\, \boldsymbol{F}\, \boldsymbol{S}\, \boldsymbol{F}^{\mathrm{T}}, \qquad \boldsymbol{S} = 2\, \frac{\partial W}{\partial \boldsymbol{C}}, \quad W \text{ strain energy density function.}$$

0D scalar reduction ($C = \lambda^2$, $J = 1$), the σ_θ in Laplace (1)

$$\sigma = \tfrac{1}{J}\, \lambda\, S\, \lambda = \lambda^2 S = 2\lambda^2\, \frac{\partial W}{\partial C} = \lambda\, \frac{\mathrm{d}W}{\mathrm{d}\lambda}.$$

Hyperelasticity: Stress from a Stored-Energy Function

A **hyperelastic** material stores energy W (deformation is path-independent).

3D energy $W(\boldsymbol{C})$, stress $\boldsymbol{S} = 2\,\partial W/\partial \boldsymbol{C}$

$$W_{\text{neo-Hooke}} = \tfrac{c}{2}\,(I_1 - 3), \qquad W_{\text{Fung}} = \tfrac{c_1}{4c_2}\left[e^{\,c_2(I_4-1)^2} - 1\right]$$

Invariants of $\boldsymbol{C}$: $I_1 = \operatorname{tr}\boldsymbol{C}$ (isotropic, elastin); $I_4 = \boldsymbol{M}\cdot\boldsymbol{C}\,\boldsymbol{M} = \lambda_{\text{fib}}^2$, the squared stretch along the reference fiber direction $\boldsymbol{M}$ (anisotropy, collagen).

0D reduction used here, one stretch λ, scalar $\sigma = \lambda\,\mathrm{d}W/\mathrm{d}\lambda$

$$\sigma_{\text{neo-Hooke}} = c\,(\lambda^2 - \lambda^{-1}), \qquad \sigma_{\text{Fung}} = c_1\,\lambda^2(\lambda^2 - 1)\,e^{\,c_2(\lambda^2-1)^2} \tag{7}$$

constituent laws as in Latorre, Humphrey, Comput Methods Appl Mech Eng (2020)

Kinematic Growth

Multiplicative Decomposition: Growth vs. Elasticity

Split the deformation into a stress-free **growth** part $\boldsymbol{F}_g$ and a stress-carying **elastic** part $\boldsymbol{F}_e$.

3D a product of tensors

$$\boldsymbol{F} = \boldsymbol{F}_e\,\boldsymbol{F}_g, \qquad \text{only } \boldsymbol{F}_e \text{ stores energy.}$$

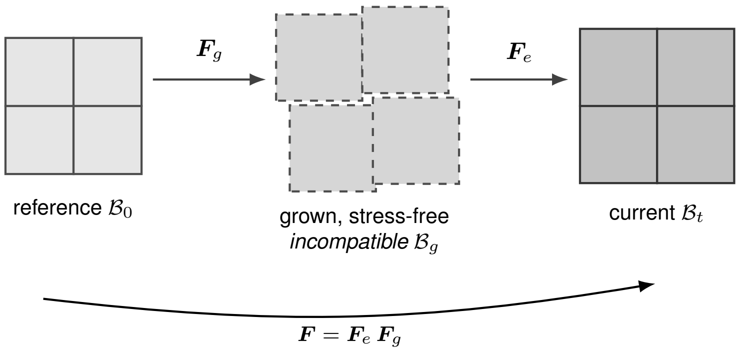
0D scalar stretches

$$\lambda = \lambda_e\,\lambda_g \tag{8}$$

Rodriguez, Hoger, McCulloch, J Biomech (1994)

Multiplicative Decomposition: Growth vs. Elasticity

$\boldsymbol{F}_g$ yields an *incompatible* stress-free state, $\boldsymbol{F}_e$ restores compability via residual stress.



Kinematic Growth Tensor

$\boldsymbol{F}_g$ encodes *how* the tissue is assumed to add mass. A common choice grows the tissue an amount θ along a fixed direction $\boldsymbol{f}$:

3D prescribed growth tensor

$$\boldsymbol{F}_g = \boldsymbol{I} + (\theta - 1)\, \boldsymbol{f} \otimes \boldsymbol{f} \;=\; \mathrm{diag}(1,\, 1,\, \theta) \tag{9}$$

0D growth stretch

$$\lambda_g \quad \text{(scalar growth stretch)}, \qquad \theta = \det \boldsymbol{F}_g = M/M_0.$$

Rodriguez, Hoger, McCulloch, J Biomech (1994)

Growth Law

The growth *pattern* $\boldsymbol{F}_g$ is typically fixed. Its rate $\dot{\boldsymbol{F}}_g = \dot{\theta}\, \boldsymbol{f} \otimes \boldsymbol{f}$ reduces to *one* scalar ODE:

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = k_g\, \theta \left(\frac{\bar{\sigma}}{\bar{\sigma}_h} - 1 \right) \tag{10}$$

Since $k_g\, \theta > 0$, the sign of the response follows the stress deviation:

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} \begin{cases} > 0 & \bar{\sigma} > \bar{\sigma}_h \quad \text{(growth)} \\ = 0 & \bar{\sigma} = \bar{\sigma}_h \quad \text{(homeostasis)} \\ < 0 & \bar{\sigma} < \bar{\sigma}_h \quad \text{(atrophy)} \end{cases}$$

Rodriguez, Hoger, McCulloch, J Biomech (1994)

Getting Started with the Code

One runner drives *every* exercise: you never edit Python, just point it at a small input file. **First-time setup** (needs Python ≥ 3.10; dependencies: NumPy, SciPy, Matplotlib, PyYAML):

```
git clone https://github.com/YaleCBL-Teaching/Growth-Remodeling
cd Growth-Remodeling
```

uv (recommended):

```
uv sync
```

pip (your own Python):

```
python -m venv .venv
pip install -e .
```

What is where:

- `run.py`: the single runner for all exercises
- `configs/`: one YAML input file per exercise *(you edit these)*
- `src/gr/`: the models & maths (read, don't edit)
- `docs/`: documentation
- `slides/`: semianr notes

Setup: Example (hypertensive artery, `ex02`)

Example. A pressurised artery is made *hypertensive* by a sustained pressure step. Kinematic growth thickens the wall. Does the wall stress $\bar{\sigma}/\bar{\sigma}_h$ return to the *prescribed* set-point?

Running an Exercise

Point the runner at the input file (`uv run`, or plain `python` once the venv is activated):

```
uv run python run.py configs/ex02_kinematic_growth.yaml
```

You get: the key numbers ($\bar{\sigma}/\bar{\sigma}_h$, mass, `diverged`) print to the terminal, and the time history opens as a plot.

Change one value: edit the YAML, or override on the command line:

```
uv run python run.py configs/ex02_kinematic_growth.yaml -set insult.pressure_factor=2
```

Parametric study: one curve per value:

```
...  -sweep insult.pressure_factor=1.5,2,3
```

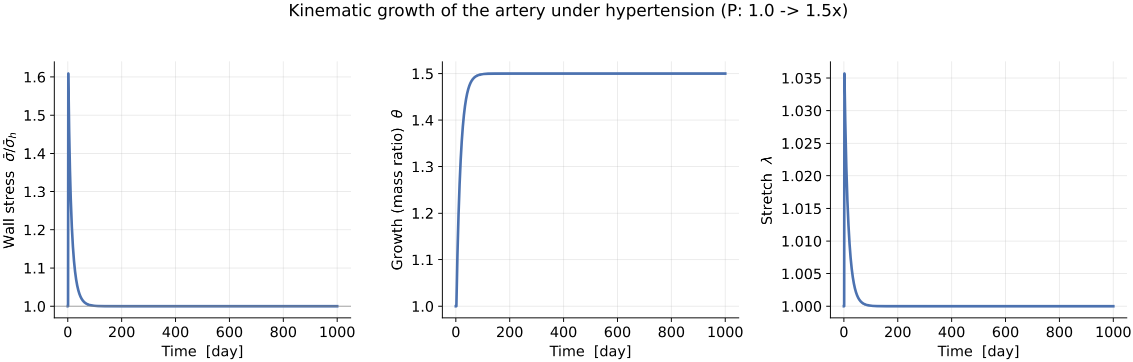
Your turn: Kinematic growth

Question. Does the wall stress return to the prescribed set-point? Try larger pressure steps. Does the outcome ever change?

Hints. $\bar{\sigma}/\bar{\sigma}_h$ settles at 1 while θ climbs to the pressure factor. Under pure hypertension it *always* adapts. Runaway needs the mixture / aneurysm story (later).

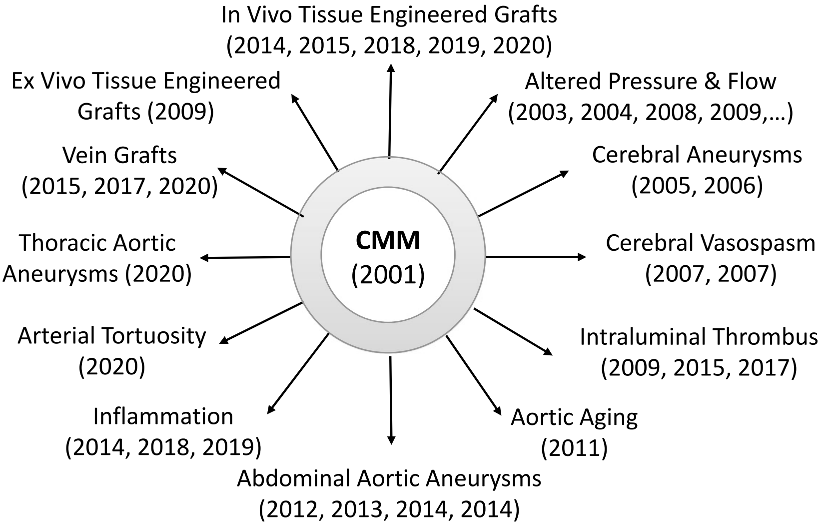
Answer: Kinematic growth

The stress plateau *always* lands on the prescribed set-point. The wall simply thickens more for a larger pressure step. Kinematic growth *imposes* stability; it cannot predict runaway.



The Constrained Mixture Model

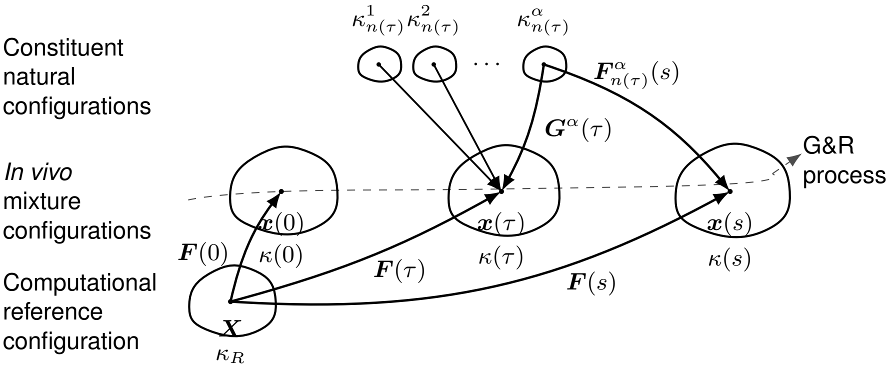
One Framework, Many Applications



Ubiquitous Vascular G&R

Humphrey, J Elasticity (2021)

The Idea: Track Constituents Over Time



κ: configuration • τ: deposition time • s: current time • α: constituent • $\boldsymbol{G}^\alpha$: deposition stretch • $\boldsymbol{F}^\alpha_{n(\tau)}$: constituent deformation

Load-Bearing Vessel Constituents

The wall is a mixture of three constituents, each with a reference **mass fraction** ϕ_0^α ($\sum_\alpha \phi_0^\alpha = 1$) and a **deposition stretch** G^α, the prestretch cells lay it down at, which *defines* its homeostatic stress $\sigma_h^\alpha = \sigma^\alpha(G^\alpha)$; the mixture set-point is $\bar{\sigma}_h = \sum_\alpha \phi_0^\alpha \sigma_h^\alpha$.

Constituent	**Role**	ϕ_0	G^α	**law; turnover** $1/k_d$
Elastin	bears load at low stretch	0.30	1.40	neo-Hooke; **none** (decades)
Collagen	stiff, crimped; anti-rupture	0.35	1.08	Fung; fast (∼80 d)
Smooth muscle	passive + active; the sensor	0.35	1.20	Fung; fast (∼80 d)

Elastin cannot be renewed ($k_d^e = 0$), its loss drives *aneurysm*; collagen & SMC **turn over**, so the wall can *adapt*.

Every Cohort Shares One Stretch

A cohort of α deposited at τ is laid down at G^α and then stretches *with the mixture*: every constituent and cohort is **constrained** to the same tissue stretch $\lambda(s)$.

3D constituent-specific deformation

$$\boldsymbol{F}^{\alpha}_{n(\tau)}(s) = \boldsymbol{F}(s)\,\boldsymbol{F}^{-1}(\tau)\,\boldsymbol{G}^{\alpha}(\tau).$$

0D scalar elastic stretch

$$\lambda^{\alpha}_{n(\tau)}(s,\tau) = \frac{\lambda(s)}{\lambda(\tau)}\,G^{\alpha} \tag{11}$$

Mass Balance

Track each constituent's **apparent mass density** ρ^α per current mixture volume, $\rho = \sum_\alpha \rho^\alpha$.

Spatial mass balance (one per constituent):

$$\frac{\partial \rho^\alpha}{\partial s} + \underbrace{\mathrm{div}(\rho^\alpha \boldsymbol{v}^\alpha)}_{\text{transport}} = \underbrace{\bar{m}^\alpha}_{\text{net mass exchange}}, \qquad \alpha = 1, \ldots, N.$$

The net rate is *signed*:

$$\bar{m}^\alpha = \underbrace{m^\alpha}_{\text{production}\geq 0} - \underbrace{r^\alpha}_{\text{removal}\geq 0} \begin{cases} < 0 & \text{atrophy} \\ = 0 & \text{maintenance} \\ > 0 & \text{growth} \end{cases}$$

From Balance to a Heredity Integral

Assumption 1 (constrained mixture): every constituent moves with the mixture, $\boldsymbol{v}^\alpha = \boldsymbol{v}$.

Assumption 2 (prescribed rates): a true production rate $m^\alpha(\tau) \geq 0$ and a survival fraction $q^\alpha(s,\tau) \in [0,1]$. Integrating from the distant past to s:

$$\rho^\alpha(s) = \underbrace{\rho^\alpha(0)\, Q^\alpha(s)}_{\text{survivors of original material}} + \underbrace{\int_0^s m^\alpha(\tau)\, q^\alpha(s,\tau)\, \mathrm{d}\tau}_{\text{deposited at } \tau, \text{ still present at } s} \tag{12}$$

with $Q^\alpha(0) = 1$. **Assumption 3:** bulk density is nearly constant, $\rho(s) = \rho(0)$, mass changes track volume changes.

Linear Momentum Balance

A rule-of-mixtures for the stresses (no body forces) satisfies the classical linear-momentum balance:

$$\operatorname{div}\boldsymbol{\sigma} = \rho\,\boldsymbol{a} \quad \xrightarrow{\text{quasi-static}} \quad \operatorname{div}\boldsymbol{\sigma} = \boldsymbol{0}.$$

Quasi-static assumption: G&R evolves slowly at every time s, so the inertia $\rho\boldsymbol{a}$ is negligible.

Constitutive Relations: Cauchy Stress

In finite elasticity the Cauchy stress follows from a stored-energy function W (per unit reference mixture volume):

$$\boldsymbol{\sigma}(s) = \frac{2}{J}\,\boldsymbol{F}(s)\,\frac{\partial W(s)}{\partial \boldsymbol{C}(s)}\,\boldsymbol{F}^{\mathrm{T}}(s), \qquad \boldsymbol{C} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}, \quad J = \det \boldsymbol{F}.$$

Rule of mixtures: $W = \sum_\alpha \hat{W}^\alpha$. Closing the model needs **three constituent-specific relations**:

- a *stored energy* W^α,
- a *production* rate m^α, and
- a *removal* (survival) q^α.

Three Constituent-Specific Constitutive Relations

1. Material: constituent stored energy W^α (e.g., neo-Hookean, Fung-type)
2. Rate of production: up with wall stress and inflammation, down with wall shear:

$$m^\alpha(\tau) = m_o^\alpha\big(1 + \underbrace{K_\sigma^\alpha\,\Delta\sigma}_{\text{circumferential stress}} - \underbrace{K_{\tau_w}^\alpha\,\Delta\tau_w}_{\text{wall shear stress}} + \underbrace{K_\varphi^\alpha\,\Delta\rho_\varphi}_{\text{inflammation}}\big) \geq 0 \tag{13}$$

3. Rate of removal: a survival fraction that decays faster under stress perturbation:

$$q^\alpha(s,\tau) = \exp\Big(-\int_\tau^s k_o^\alpha\big(1 + \omega\,\Delta\sigma^2(t) + \cdots\big)\,\mathrm{d}t\Big) \in [0,1] \tag{14}$$

Constituent Stored-Energy Functions

Each cohort carries its *own* stored energy W^α. Summing over cohorts with the *same* heredity structure gives the apparent (mixture-level) energy:

$$\rho\,\hat{W}^\alpha(s) = \underbrace{\rho^\alpha(0)\,Q^\alpha(s)\,W^\alpha\big(\boldsymbol{F}^\alpha_{n(0)}(s)\big)}_{\text{original material}} + \underbrace{\int_0^s m^\alpha(\tau)\,q^\alpha(s,\tau)\,W^\alpha\big(\boldsymbol{F}^\alpha_{n(\tau)}(s)\big)\,\mathrm{d}\tau}_{\text{later cohorts}} \tag{15}$$

, with elastic deformation

$$\boldsymbol{F}^\alpha_{n(\tau)}(s) = \boldsymbol{F}(s)\,\boldsymbol{F}^{-1}(\tau)\,\boldsymbol{G}^\alpha(\tau) \tag{16}$$

. The mixture energy $W = \sum_\alpha \hat{W}^\alpha$ closes the mixture stress.

Special Cases: Before Perturbation and Homeostasis

(I) At $s = 0$ (just before a perturbation) only original material is present, and $Q^\alpha(0) = 1$:

$$\rho\,\hat{W}^\alpha(0) = \rho^\alpha(0)\,W^\alpha\big(\boldsymbol{F}^\alpha_{n(0)}(0)\big) \;\Rightarrow\; \hat{W}^\alpha(0) = \underbrace{\tfrac{\rho^\alpha(0)}{\rho(0)}}_{\phi^\alpha(0)}\,W^\alpha(0),$$

recovering the rule of mixtures $W(0) = \sum_\alpha \phi^\alpha(0)\,W^\alpha(0)$.

(II) Homeostasis: turnover in an *unchanging* configuration, so $\boldsymbol{F}^\alpha_{n(\tau)}(s) = \boldsymbol{F}^\alpha_{n(0)}(s)$ (independent of τ). Then W^α factors out and, using the mass balance,

$$\rho\,\hat{W}^\alpha(s) = \underbrace{\left(\rho^\alpha(0)Q^\alpha(s) + \int_0^s m^\alpha q^\alpha\,\mathrm{d}\tau\right)}_{=\,\rho^\alpha(s)}\,W^\alpha\big(\boldsymbol{F}^\alpha_{n(0)}(s)\big) \;\Rightarrow\; \hat{W}^\alpha(s) = \phi^\alpha(s)\,W^\alpha(s). \tag{17}$$

Perfect Maintenance

Take *constant* rates: $m^\alpha = m_0$, $Q^\alpha(s) = e^{-K^\alpha s}$, $q^\alpha(s,\tau) = e^{-K^\alpha (s-\tau)}$. The mass balance integrates to

$$\rho^\alpha(s) = \underbrace{\rho^\alpha(0)\, e^{-K^\alpha s}}_{\text{original decays}} + \underbrace{\frac{m_0}{K^\alpha}\left(1 - e^{-K^\alpha s}\right)}_{\text{production accumulates}} .$$

A time-*invariant* density ($\rho^\alpha(s) = \rho^\alpha(0)$ for all s) requires the two to balance exactly:

$$\text{perfect maintenance} \iff \frac{m_0}{K^\alpha} = \rho^\alpha(0). \tag{18}$$

Production m_0 then exactly offsets removal $K^\alpha \rho^\alpha$ (continuous turnover, unchanging tissue).

Your turn: Computational Cost?

Question. At each of N time steps the mixture stress sums over *all* cohorts deposited so far. How does the total cost scale with N?

Hints. Step k re-sums $\sim k$ stored cohorts; add up $\sum_{k=1}^{N} k$.

Your turn: Computational Cost?

Question. At each of N time steps the mixture stress sums over *all* cohorts deposited so far. How does the total cost scale with N?

Hints. Step k re-sums $\sim k$ stored cohorts; add up $\sum_{k=1}^{N} k$.

Answer: Computational Cost?

Step k re-sums the $\sim k$ cohorts stored so far, so the total work is

$$\sum_{k=1}^{N} k = \frac{N(N+1)}{2} = \mathcal{O}(N^2)$$

quadratic in the number of time steps. The *homogenized* model (next) removes this history sum, cutting the cost to $\mathcal{O}(N)$.

Setup: Example 2: Turnover under Hypertension (`ex03`)

Example. The same artery, now with collagen & SMC turning over (elastin does not). Which constituent grows to carry the load, and does turnover restore the set-point? Equilibrium each step: $\bar{\sigma}(\lambda) = \bar{\sigma}_h \, \gamma \, \lambda^2 / m$ (load factor γ, mass ratio m).

Run it.

```
uv run python run.py configs/ex03_constrained_mixture.yaml
```

Sets each constituent's `turnover_days` $(1/k_d)$, `gain` (K_σ), and the `insult`, hypertension (γ) or aneurysm (`elastin_surviving`).

Parametric study.

```
...  -set model.constituents.collagen.gain=3.0
```

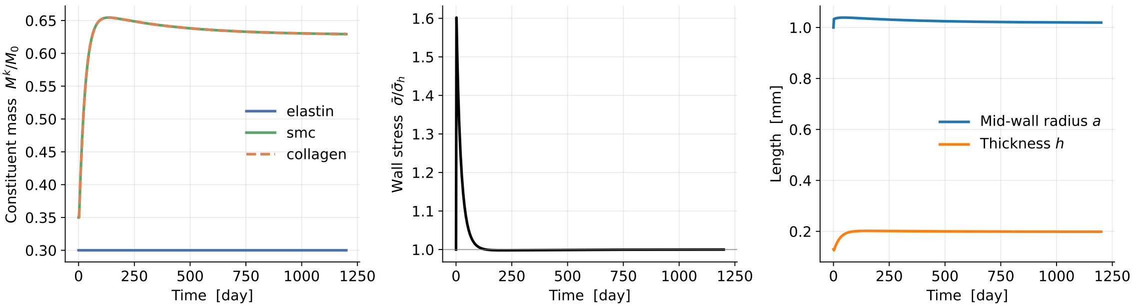
Your turn: Constrained Mixture Model

Question. Does the same tissue *adapt* to hypertension but *run away* on the aneurysm insult? Which constituent grows to carry the load elastin used to bear?

Answer: Constrained Mixture Model

Mild hypertension **adapts**: collagen and SMC grow, tissue stress returns to homeostatic. A severe elastin insult can **run away**. Faster turnover or higher gain reach the new state sooner.

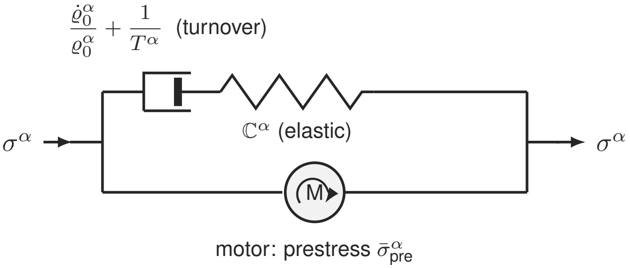
Full constrained-mixture model: adaptation to hypertension (artery)



Homogenized CMM

The Mechanical Analog: A Maxwell Fluid with a Motor

Each constituent behaves like a **Maxwell element in parallel with a motor**: turnover relaxes stress toward a *prestress*, not zero.



σ^α	constituent stress (the load)
$\mathbb{C}^\alpha$	elastic stiffness (spring)
$\dot{\varrho}_0^\alpha/\varrho_0^\alpha + 1/T^\alpha$	turnover + degradation (dashpot)
$\bar{\sigma}^\alpha_{\text{pre}}$	deposition prestress (motor)

Back to the full CMM (0D):

- spring $\leftrightarrow$ constituent law $\sigma^\alpha(\lambda_e^\alpha)$
- dashpot $\leftrightarrow$ mass turnover / removal k_d^α
- motor $\leftrightarrow$ deposition stretch G^α

Cyron, Aydin, Humphrey, Biomech Model Mechanobiol (2016)

Growth (Mass) and Remodeling (Natural Configuration)

Two 0D ODEs per constituent replace the cohort history. **Growth**: mass production $\propto$ mass, driven by the tissue-stress deviation. **Remodeling**: new mass enters at natural stretch λ/G^α, old mass leaves at the current λ_n^α; mixing the two at the turnover rate evolves the mean.

0D growth (mass)

$$\frac{\mathrm{d}M^\alpha}{\mathrm{d}t} = M^\alpha K_\sigma^\alpha k_d^\alpha \left(\frac{\bar{\sigma}}{\bar{\sigma}_h} - 1\right) \tag{19}$$

0D remodeling (mean natural stretch)

$$\frac{\mathrm{d}\lambda_n^\alpha}{\mathrm{d}t} = \frac{m^\alpha}{M^\alpha}\left(\frac{\lambda}{G^\alpha} - \lambda_n^\alpha\right) \tag{20}$$

with production $m^\alpha = k_d^\alpha M^\alpha \Upsilon^\alpha$. Holding λ fixed, remodeling relaxes each constituent's stress to its prestress σ_h^α (time constant $\sim 1/k_d$), the Maxwell-with-motor behaviour.

Your turn: Why Is the Homogenized Model Cheap?

Question. How does the cost of the homogenized constrained mixture model scale with the number of steps N, compared with the full model?

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Answer: Why Is the Homogenized Model Cheap?

The state is *fixed-size*, two numbers $(M^\alpha, \lambda_n^\alpha)$ per constituent, and each step just advances two ODEs, with *no* history to re-sum:

$$\text{homogenized: } \mathcal{O}(N) \qquad \text{vs.} \qquad \text{full CMM: } \mathcal{O}(N^2)$$

Same time trajectory, far cheaper, and the fixed state maps onto a viscoelastic (Maxwell-with-motor) material, so it drops traight into standard finite-element solvers.

Your turn: When do homogenized and full agree?

Question. The homogenized model keeps only the *mean* natural configuration per constituent. Under what assumptions should it match the full deposition-history model and when would you expect them to diverge?

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Answer: When do homogenized and full agree?

They agree when each constituent's cohorts share *nearly the same* natural configuration i.e. slow, gradual loading where deposition stretch G^α and turnover are roughly stationary, so the cohort spread stays narrow and the mean represents it well. **They diverge** under fast or strongly time-varying insults, when cohorts deposited at very different states coexist and a single mean stretch can no longer capture the distribution.

Setup: Example (homogenized vs. full, `ex04`)

Example. Run both mixture theories on the *same* artery and insult. How closely does the homogenized model track the full model, and how much faster is it?

Run it.

```
uv run python run.py configs/ex04_homogenized_vs_full.yaml
```

Here `study: compare` runs `theories: [constrained_mixture, homogenized_cmm]` on one scenario and prints each runtime and the largest mass-ratio gap between them.

Parametric study. Stress the comparison with a harsher, longer insult:

```
...  -set insult.elastin_surviving=0.3 -set insult.ramp=10
...  -set simulate.t_end=4000     # does the agreement hold?
```

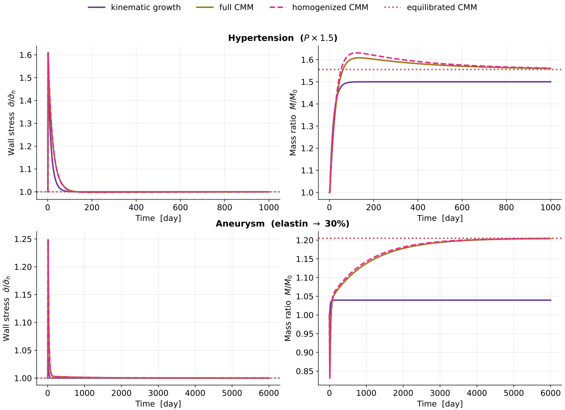
Your turn: Homogenized vs. full

Question. Is the difference between the two curves visible by eye? By what factor is the homogenized model faster and does the agreement survive a harsher, longer insult?

Hints. Read the printed mass-ratio gap and the two runtimes. The homogenized model should track the full one closely wherever the loading is slow and gradual.

Answer: Homogenized vs. full

The homogenized curve overlays the full CMM closely at a fraction of the runtime ($O(N)$ vs. $O(N^2)$). **Learning:** it reproduces a **similar time trajectory** to the full model, cheaply.



Equilibrated CMM

Directly Solve for the End State

Set every rate to zero and solve the **algebraic** equilibrium, one root-find instead of thousands of time steps.

3D the 3D equilibrium conditions

$$\underbrace{\Upsilon = 1 \Rightarrow \bar{\sigma}^* = \bar{\sigma}_h}_{\text{(A) growth balance}} \qquad \underbrace{\boldsymbol{F}_n^\alpha = \text{const} \Rightarrow \sigma^\alpha = \sigma_h^\alpha}_{\text{(B) remodeling completed}} \qquad \underbrace{\nabla \cdot \boldsymbol{\sigma} = \boldsymbol{0}}_{\text{(C) mechanical equilib.}}$$

0D scalar conditions at the evolved geometry

$$\bar{\sigma}^* = \bar{\sigma}_h, \qquad \lambda_e^\alpha = G^\alpha, \qquad \sigma_\theta = \frac{Pa}{h}.$$

Collapse to One Scalar Equation

Let the turnover constituents grow in fixed proportion (exact when they share gain and turnover). The 3D conditions (A)–(C) collapse to a *single* equation for the evolved stretch λ^*, consistent with the same $\lambda,\ G^\alpha,\ \phi^\alpha$ used throughout:

0D scalar equilibrium equation for λ^*

$$\phi^e\, s + s\,\frac{\bar{\sigma}_h - \sigma^e(G^e\lambda^*)}{\sigma_{\text{turn}} - \sigma_h^e} - \gamma\,(\lambda^*)^2 = 0 \tag{21}$$

with the Laplace exponent $n = 2$, γ the sustained load factor, s the surviving elastin fraction, ϕ^e the elastin mass fraction. Once λ^* is known, the evolved mass ratio is $M/M_0 = \gamma(\lambda^*)^2$.

This equation *is* the fixed point of the homogenized ODEs ((19)–(20) with $\mathrm{d}/\mathrm{d}t = 0$), so its root coincides with the long-time limit of the transient theory.

The Punchline: Existence = Stability

Equation (21) may have **no physical root**. Then no adapted state exists and the transient theories dilate without bound. The solver is therefore the cleanest **stability test**: root $\Rightarrow$ adapts, no root $\Rightarrow$ unstable. Near the boundary the transient slows critically, which is exactly why the instant algebraic test is so useful.

Setup: Example (the equilibrated stability test, `ex05`)

Example. Skip the transient: solve the single algebraic equation (21) for the *final* adapted stretch λ^*. A physical root means the tissue adapts; *no root* means it dilates without bound. Scanning elastin loss then traces the stability boundary.

Run it.

```
uv run python run.py configs/ex05_equilibrated.yaml
```

Here `study: equilibrium` solves once for the given insult, then the `scan:` block sweeps `elastin_surviving` and `report_critical: elastin` prints the critical surviving fraction s_{crit}.

Parametric study. Move the boundary by stiffening the wall, then re-read s_{crit}:

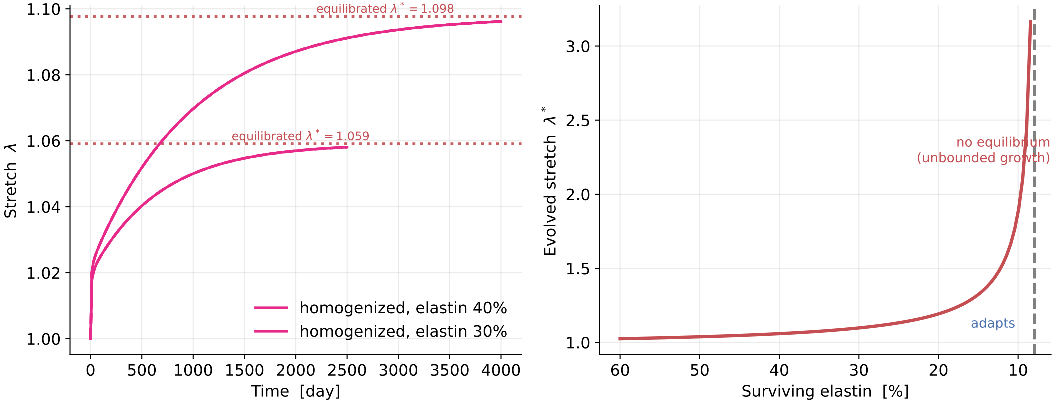
```
...  -set model.constituents.collagen.gain=3.0
...  -set model.constituents.collagen.law.c1=600
```

Your turn: Equilibrated

Question. Does the algebraic λ^* land on the transient plateau where a root exists? What is the critical elastin loss beyond which *no* equilibrium exists at all?

Answer: Equilibrated

Where a root exists, the equilibrated solve lands exactly on the transient end-state. Below s_{crit} there is *no* root: the transient dilates without bound.

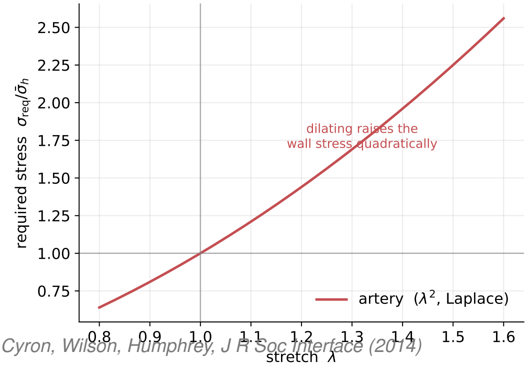


Stability

Where Instability Comes From: The Laplace Exponent

In a pressurised tube the wall stress is $\bar{\sigma} = \frac{Pa}{h}$ with radius $a \propto \lambda$ and thickness $h \propto M$, so the required (Laplace) stress is *quadratic* in the stretch, $\bar{\sigma}_{\text{req}} \propto \gamma\,\lambda^2/M$. The homeostatic state is stable only if adding mass *lowers* the stress it must carry:

$$\left.\frac{\mathrm{d}\bar{\sigma}}{\mathrm{d}\theta}\right|_{\theta^*} < 0 \iff \text{tissue adapts} \tag{22}$$



Growth thickens the wall ($h \propto M$), which lowers stress; but dilation raises it as λ^2 (Laplace). While enough **elastin** survives, its stiff recoil keeps $\mathrm{d}\bar{\sigma}/\mathrm{d}\theta < 0$ and the artery **adapts**. Lose enough elastin and the sign flips, $\mathrm{d}\bar{\sigma}/\mathrm{d}\theta > 0$: **runaway**. It is the quadratic Laplace exponent that makes this possible.

Cyron, Wilson, Humphrey, J R Soc Interface (2014)

Same Tissue, Two Insults

Read stability two ways: **existence**: the equilibrated equation (21) has a root iff the tissue can adapt (no time integration); or **transient**: integrate and watch it plateau or climb. The same tissue can do either: mild elastin loss adapts, severe loss dilates without bound.

Setup: Example (the stable/unstable map, `ex06`)

Example. A tissue *adapts* to an insult iff a mechanobiological equilibrium exists (21). Sweeping that instant test over (surviving elastin) $\times$ (pressure) draws the stable/unstable map. Then change *one* tissue property and watch the unstable "aneurysm" region shrink.

Run it.

```
uv run python run.py configs/ex06_stability.yaml
```

Here `study: map` builds the 2D grid over the two `axes:` via the `adapts` existence test and shades adapt (blue) vs. runaway (red); it also prints the fraction of the grid that adapts.

Parametric study. Change one property in the `model:` block (gain, collagen `c1`, or G), e.g.

```
... -set model.constituents.collagen.gain=3.0
... -set model.constituents.collagen.G=1.15
```

Then compare how much the stable region grows.

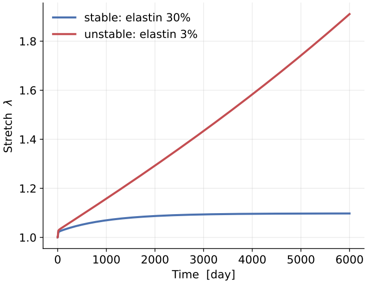
Your turn: Stability map

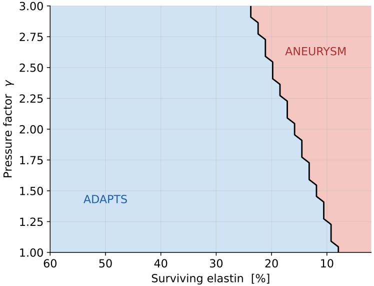
Question. Which single change (higher gain, stiffer collagen, or larger deposition stretch) enlarges the stable region the most? Elastin cannot be regrown in adults: what does that imply for aneurysm?

Hints. Compare the printed “fraction that adapts” across the three changes. Each shifts the boundary between adaptation and runaway.

Answer: Stability map

Higher gain, stiffer collagen, or larger deposition stretch all enlarge the *adapts* region, the tissue tolerates a stronger insult before running away.

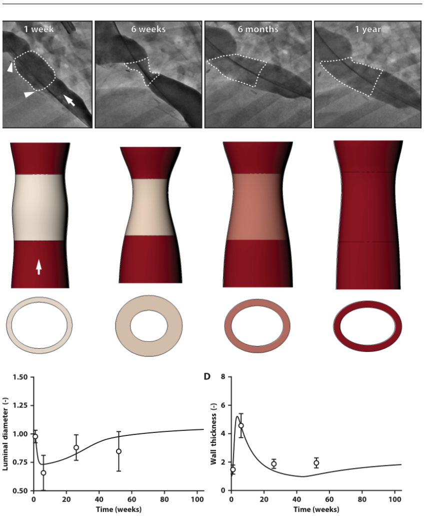




Learning: the existence test that defines the equilibrium boundary *is* the stability criterion.

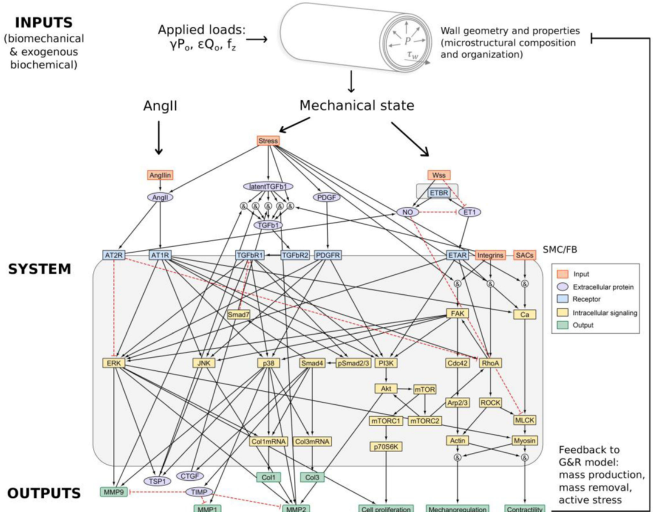
Future Perspectives

Clinical Application: Reversible Graft Stenosis



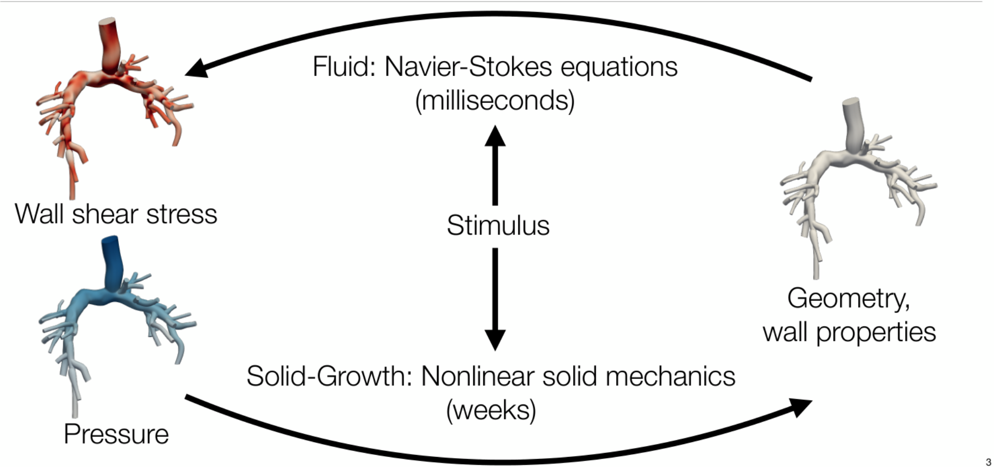
Drews, Pepper, Best, . . . , Humphrey, Breuer, Sci Transl Med (2020)

Systems-Biology Coupling: Signaling to Remodeling



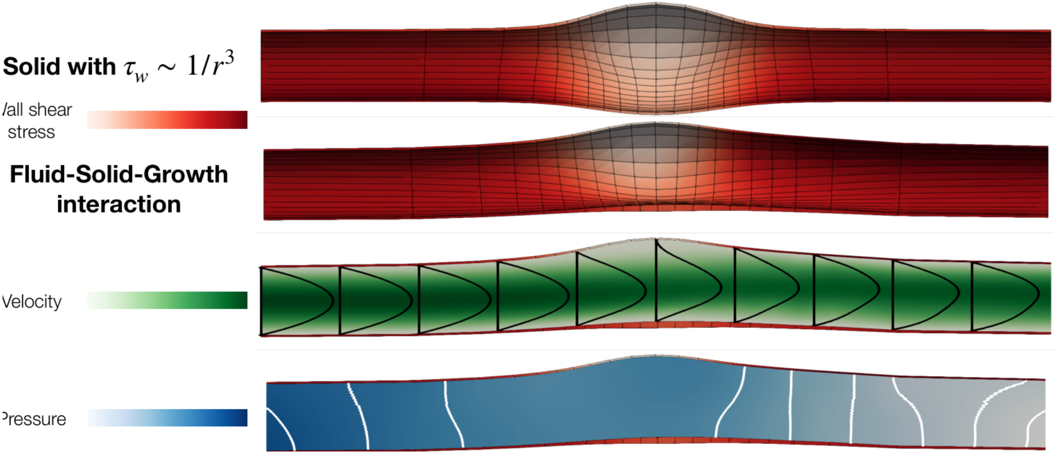
Irons, Humphrey, PLoS Comput Biol (2020); Irons, Latorre, Humphrey, Ann Biomed Eng (2021)

Fluid–Solid–Growth: Closing the Hemodynamic Loop



Schwarz, Pfaller, . . . , Marsden, Comput Methods Appl Mech Eng (2023);
Pfaller, Latorre, . . . , Marsden, Comput Methods Appl Mech Eng (2024)

FSGe: Long-Term Evolved Aneurysmal Equilibrium



Open-Source Implementations

Under the Hood, What Each File Computes (1/2: the core)

The `gr` package is layered: a shared **continuum-mechanics core** (read, don't edit), then one file per G&R theory.

File	Computes	Continuum mechanics / G&R
`mechanics.py`	strain energy $W(\lambda_e)$, 2nd PK stress $S = \mathrm{d}W/\mathrm{d}E_e$, Cauchy $\sigma = \lambda_e^2 S$ (neo-Hookean, Fung)	finite-strain **hyperelasticity** of each constituent, in reference measures
`parameters.py`	`Model`/`Constituent` $(\phi_0, G^\alpha, k_d, K_\sigma)$; set-point $\sigma_h^\alpha = \sigma^\alpha(G^\alpha)$; `Insult`	**deposition prestretch** & homeostatic **set-points**; the two insults
`geometry.py`	required Laplace stress $\bar{\sigma}_h\,\gamma\,\lambda^2/m$; solves $\bar{\sigma}(\lambda) = \sigma_{\mathrm{req}}$ for λ	**mechanical equilibrium**; no root $\Rightarrow$ runaway
`history.py`	`Result`: $t, \lambda, M/M_0, \bar{\sigma}, \ldots$	shared time-history output

Under the Hood, What Each File Computes (2/2: the theories)

File	Computes	G&R theory / role
`kinematic_growth.py`	split $\boldsymbol{F} = \boldsymbol{F}_e \boldsymbol{F}_g$, scalar θ; $\dot{\theta} = k_g \theta(\bar{\sigma}/\bar{\sigma}_h - 1)$	Theory 1: kinematic growth
`constrained_mixture.py`	heredity integrals over cohorts; mass balance + mixture stress ($O(N^2)$)	Theory 2: full CMM
`homogenized_cmm.py`	two ODEs per constituent: $\dot{M}^\alpha$, $\dot{\lambda}_n^\alpha$ ($O(N)$)	Theory 3: homogenized CMM
`equilibrated_cmm.py`	rates $\to 0$: one algebraic solve for λ^*	Theory 4: equilibrated CMM
`stability.py`	`adapts` (does a root exist?), `critical_elastin_loss`	stability $=$ **existence** of an equilibrium
`scenario.py`, `run.py`, `configs/`	build model/geometry/insult from YAML; run a study; sweep	the exercise driver (input files)

Common thread: each theory changes only *how the supplied stress $\bar{\sigma}(\lambda)$ is built*, the same constituent laws and equilibrium solve are reused, so every difference you saw is the G&R law alone.

Research-Grade Open-Source Solvers

Project	What it does	Stack / group
svGrowth	constrained-mixture G&R of vessels (biaxial thin-wall); the cross-validation reference used above	Python; Stanford CBCL
svFSGe	fast, strongly-coupled 3D fluid–solid–growth driver (the FSGe method)	Python + svFSI/svMultiPhysics; Stanford CBCL
FEMBE_Plugin	mechanobiologically equilibrated constrained-mixture material as a FEBio plugin	C++ / FEBio; Yale (Humphrey lab)
FEFSG_Plugin	fluid–solid–growth coupling as a FEBio plugin (thoracic-aorta hypertension & aneurysm cases)	C++ / FEBio; Yale (Humphrey lab)

`github.com/StanfordCBCL/svGrowth` `github.com/StanfordCBCL/svFSGe`

`github.com/yale-humphrey-lab/FEMBE_Plugin` `github.com/yale-humphrey-lab/FEFSG_Plugin`

svGrowth builds on Humphrey, Rajagopal, Math Models Methods Appl Sci (2002); FEMBE on Latorre, Humphrey, Comput Methods Appl Mech Eng (2020)

How the Four Theories Answer: The Takeaways

Theory	Stability?	What it says
Kinematic growth	*Imposed*	Either reaches the *a-priori prescribed* homeostasis, or is unstable, nothing in between. Built in, not learned.
Constrained mixture (full)	*Predicted*	Adapts or dilates **depending on the insult**, the faithful reference.
Homogenized CMM	*Predicted*	A **similar time trajectory** to the full model, at a fraction of the cost.
Equilibrated CMM	*Instantly*	**Matches** the transient theories *if* an equilibrium exists; *no* root $\Rightarrow$ it does not match, unbounded growth.

Summary: What to Take Away

The biology. Living tissue continuously **turns over**; cells deposit new material *pre-stretched* at G^α, chasing a homeostatic mechanical set-point σ_h.

The four theories, one set-point, one question: does it adapt or run away?

- **Kinematic growth**: prescribe $\boldsymbol{F}_g$; *imposes* the set-point. Outcome is binary: reach the prescribed homeostasis, or go unstable.
- **Constrained mixture**: track every cohort; stability is *predicted* and **depends on the insult** (mild $\rightarrow$ adapt, severe $\rightarrow$ runaway).
- **Homogenized**: one mean natural configuration per constituent; a **similar time trajectory** to the full model at $O(N)$ not $O(N^2)$.
- **Equilibrated**: solve for the end state directly; **matches** the transient theories *iff* an equilibrium exists, and flags instability by finding no root.

Where it goes. These tools now drive **clinical prediction** (TEVGs), **systems-biology coupling**, and **fluid–solid–growth**.

Key references I

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Key references II

- Figueroa, Baek, Taylor, Humphrey. A computational framework for fluid–solid–growth modeling. *Comput Methods Appl Mech Eng* (2009).
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- Ambrosi, Ben Amar, Cyron, . . . , Humphrey, Kuhl. Growth and remodelling of living tissues. *J R Soc Interface* (2019).

Key references III

Code

- Seminar code (this deck's figures): `github.com/YaleCBL-Teaching/Growth-Remodeling`
- `github.com/StanfordCBCL/svGrowth`, `svFSGe`
- `github.com/yale-humphrey-lab/FEMBE_Plugin`, `FEFSG_Plugin`

Figure sources. Result figures reproduced with the `gr` package; literature figures cited per-slide.